

Nidhin Mahesh A

Senior Software Engineer · Blockchain Protocols & Distributed Systems

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SUMMARY

Senior software engineer with 5+ years building blockchain protocols and distributed systems. Co-led development of the **Rubix Layer 1 protocol** (Proof-of-Pledge consensus, gasless transactions) and a **zero-knowledge rollup on zkEVM** at Jupiter Meta; shipped production self-custody wallets, staking systems, and payment infrastructure. Author of four peer-reviewed papers on blockchain security and decentralized systems.

CORE SKILLS

Languages: Golang · Rust · Solidity · Java · Python · TypeScript · Dart

Blockchain: L1 protocol design · ZK rollups / zkEVM · consensus (Proof-of-Pledge) · smart contracts · self-custody wallets · P2P networking · Hyperledger Fabric · IPFS · tokenomics · DeFi

Platform: Spring Boot · REST API design · FastAPI · Next.js · Flutter · Docker · Kubernetes · CI/CD · microservices

EXPERIENCE

Full-Stack Developer (Indie Project), Fexr · Self-employed

Dubai, UAE · Remote · Sep 2025 – Present

- Architected and deployed a production-grade **Web3 infrastructure platform**, scaling it from a single-chain custody solution to a comprehensive **multi-chain settlement layer**
- Engineered core protocol integrations — **agentic trading engines** and smart-contract systems — while managing the complete cloud-native deployment pipeline
- Own the full stack solo: resilient cloud environments, GitLab CI/CD, and custom backend services optimized for distributed execution and secure interactions with **DeFi protocols** — without the overhead of a large engineering team

Lead Engineer, Jupiter Meta

Hyderabad, India · Nov 2024 – Aug 2025

- Founding engineering team member; contributed to the design and launch of the **JMDT blockchain**, an in-house **zero-knowledge rollup on zkEVM** built to streamline internal asset settlement
- Designed **ZK circuit architectures** and proof-verification pipelines at the core of the company's product roadmap

Lead Engineer — Rubix Layer 1 Blockchain Protocol, Rubix Network

Hyderabad, India · Mar 2021 – Nov 2024

- Provided technical guidance and coordinated the engineering team that established the network's foundational infrastructure across **testnet and mainnet** environments
- Delivered the network's **first native wallet** from concept to launch — enabling initial user adoption — with **Decentralized Identifier (DID)** support via an early implementation of image-based non-linear secret sharing for secure account management
- Built the core protocol services — P2P networking, transaction processing, consensus (**Proof-of-Pledge**), and asset custody — in **Spring Boot (Java)**, exposing REST APIs for wallets, staking, and token transfers
- Led the full **migration of the protocol stack from Spring Boot to Golang** (2023), cutting memory footprint and improving runtime performance
- Drove open-source protocol development, community adoption, and developer onboarding across the gasless, energy-efficient L1 network

Research Assistant, TIFAC-CORE in Cyber Security

Coimbatore, India · Jan 2020 – Apr 2020

- Developed a secure, **permissioned blockchain** architecture establishing immutable **data provenance** and strict audit trails for a national defense organization — cryptographic integrity mechanisms, decentralized access controls, and tamper-proof record-keeping

Research Intern, Amrita Center for Wireless Networks & Applications

Kollam, India · Jun 2019 – Sep 2019

- Built a **Hyperledger** blockchain test-bed of 7 Raspberry Pi nodes running Golang smart contracts; deployed and validated P2P energy-trading workflows on constrained IoT hardware

Writer, Towards Data Science · Freelance

Remote · Dec 2017 – May 2018

- Authored machine-learning tutorials and technical guides on **TensorFlow**, translating deep-learning concepts into hands-on implementation steps for a global audience of developers and data scientists

PUBLICATIONS

Smartverse: Blockchain-Based Crowdsourced V2X Message Verification and Dissemination System — *15th Int’l Conference on Communication Systems & Networks (COMSNETS), 2023*

Blockchain Framework for Social Media DRM Based on Secret Sharing — *Springer — Int’l Conference on ICT for Intelligent Systems (ICTIS), 2021*

Conceptualizing Blockchain-Based Energy Market for Self-Sustainable Community — *ACM BlockSys Workshop @ SenSys, New York, 2019* · DOI

Decentralized Power System and Future Mobility: The Use Cases of Community-Driven Electric Vehicle Charging Infrastructure — *ACM Int’l Workshop on Smart Cities and Communities (TESCA), 2019* · DOI

EDUCATION

B.Tech, Computer Science & Engineering, Amrita School of Engineering, Amritapuri

2016 – 2020

- Led the Google Developer Student Club and contributed through the FOSS club — ranked top-10 nationally for Google Summer of Code selections